Module 5

Valuing Bonds and Stocks

FINE 3010, Financial Management
Prof. Renping Li, Tulane University

From Module 4 to Module 5

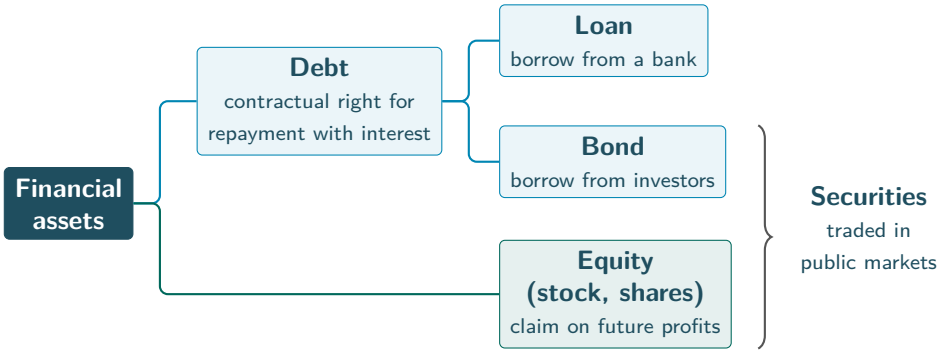
- **Module 4**, the time value of money, the present value of future cash flows

From Module 4 to Module 5

- **Module 4**, the time value of money, the present value of future cash flows
- **Module 5**, the same time-value-of-money principles value bonds and stocks
  - Bond, the present value of its coupons and face value
  - Stock, the present value of its expected dividends

Every Financial Asset Is Debt or Equity

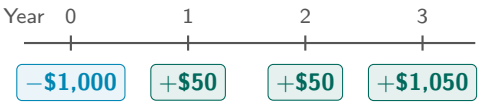
- **Financial asset**, value from a contractual right or ownership claim



Debt's Cash Flow Is Fixed, a Stock's Is Not

Debt

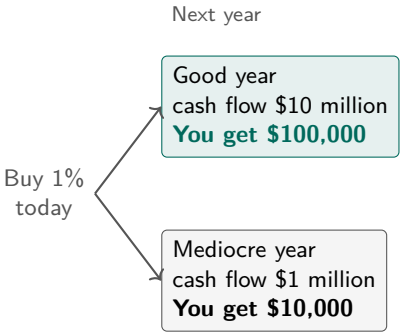
$1,000, 3 years, 5% interest



Every payment known in advance

Stock

1% of the company



Value of claim depends on how the company does

Common Types of Corporate Financing

Debt, lending to the company
Main forms, bonds and loans

Holder gets a fixed promised amount

"Senior", paid back before equity in bankruptcy

No voting rights or control

Common Types of Corporate Financing

Debt, lending to the company Main forms, bonds and loans	**Equity**, stock ownership in the company
Holder gets a fixed promised amount	Holder gets dividends
"Senior", paid back before equity in bankruptcy	"Residual claimant", paid whatever is left after all promised payments, no guaranteed payout
No voting rights or control	Voting rights

Module 5, Part 1

Bonds

Bonds, Definitions

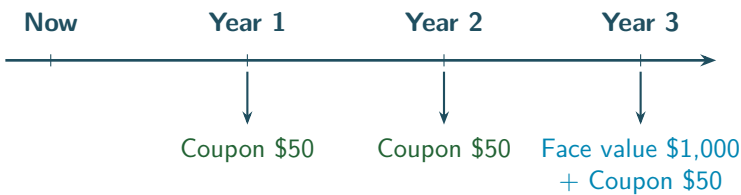
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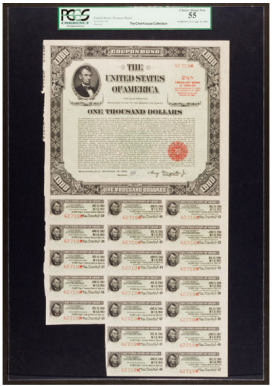
Bonds, Definitions

- **Bond**, a security that obligates the issuer (the borrower) to make specified payments to the investor (the lender, or "bondholder")
- Typically periodic payments (the "coupon"), then a "face value" at the end
- E.g., a three-year bond with an annual coupon of \$50 and a face value of \$1,000, the borrower promises to pay,



Bonds, Terminology

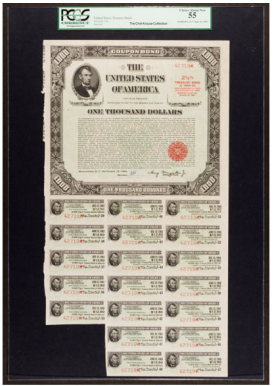
- **Face value**, the final lump-sum payment at the "maturity" of the bond, usually $1,000 per bond in the U.S.



Imaged by Heritage Auctions, HA.com

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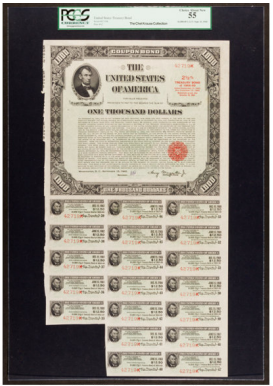
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- **Maturity**, the time when the debt expires ("matures") and the face value is paid
- **Time to maturity**, the number of years until maturity



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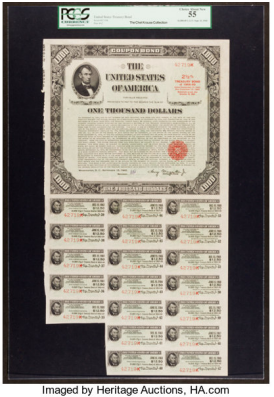


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- **Coupon**, the periodic payments to the bondholder before maturity
- **Coupon rate**,

$$\text{Coupon rate} = \frac{\text{Annual coupon}}{\text{Face value}}$$



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Bonds, Terminology

- Common fallacy, the coupon is not “interest”, and the coupon rate is not necessarily the interest rate

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- Common fallacy, the coupon is not "interest", and the coupon rate is not necessarily the interest rate
  - E.g., **Module 4**'s Puerto Rico bond, the coupon was 0 but the interest rate was 5.15%

Example, Puerto Rico Borrows

Recall from **Module 4**

- In 2007, Puerto Rico sold bonds that promised to pay $1,000 after 47 years (no periodic "coupon" payments)
- These bonds were sold at $94.40 each
- What was the interest rate paid by Puerto Rico?



Example, Puerto Rico Borrows

Recall from **Module 4**



$N = 47, \quad I = ?$

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- Solution,
  - 47 **N**
  - -94.40 **PV**
  - 0 **PMT**
  - 1000 **FV**
  - **I/Y**

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- The rate is 5.15%

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- **Bond value**, the present value of all its cash flows (coupons and face value), discounted at the "required rate of return" r
  - "Required rate of return", the name for the discount rate in this application

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- E.g., for a bond with time to maturity T,

$$\text{Value} = \text{PV} = \frac{\text{Coupon}_1}{(1+r)^1} + \frac{\text{Coupon}_2}{(1+r)^2} + \cdots + \frac{\text{Coupon}_T + \text{Face value}}{(1+r)^T}$$

Bonds, Pricing

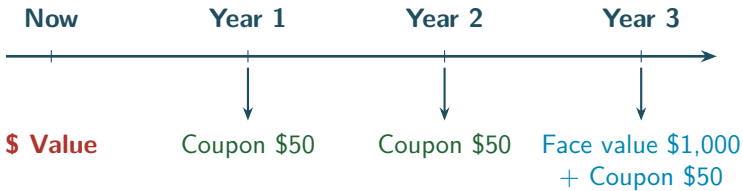
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- Investors choose the opportunity cost, "what could I get investing in something else with a similar risk", as the required rate of return r

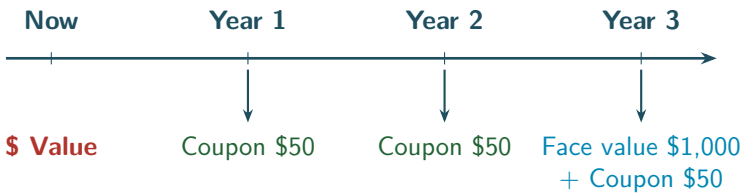
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$$\text{Value} = \frac{\$50}{(1+0.04)^1} + \frac{\$50}{(1+0.04)^2} + \frac{\$50+\$1{,}000}{(1+0.04)^3} = \$1{,}027.75$$

Example, Bond Pricing with a Financial Calculator

- The same bond, an annual coupon of $50, a face value of $1,000, three years to maturity, and a required rate of return of 4%
- Use a financial calculator to value the bond

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  - 1000 **FV**
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  - 4 **I/Y**
  - **CPT**, then **PV**

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- The output is $-\$1{,}027.75$ (why the minus sign? recall how financial calculators get to the answer in **Module 4**)

Valuing Bonds with a Financial Calculator

- The financial calculator keys map onto a bond,
  - **N**, the number of periods (years) until maturity
  - **PV**, the value of the bond (the present value of the coupons and the face value)
  - **PMT**, the coupons (in dollars)
  - **FV**, the face value (the lump sum paid at the end)
  - **I/Y**, the discount rate (the required rate of return r)

Recap of Part 1

- Companies issue bonds (and other debt) to raise financing from investors

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- Companies issue bonds (and other debt) to raise financing from investors
- In return, the companies make specified payments of coupons and face value to the investors
- The value of a bond is the discounted payments, at the required rate of return

Module 5, Part 2

Bond Yields

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- **Yield**, the discount rate y for which the present value of the bond's payments (coupons and face value) equals its price

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- **Yield**, the discount rate y for which the present value of the bond's payments (coupons and face value) equals its price
- Given the price of a bond (PV), its maturity T, and its coupons, find the yield by solving for y,

$$\text{Price} = \frac{\text{Coupon}}{(1+y)^1} + \frac{\text{Coupon}}{(1+y)^2} + \cdots + \frac{\text{Coupon} + \text{Face value}}{(1+y)^T}$$

Example, Puerto Rico Borrows

Recall from **Module 4**

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- **Here we are calculating the yield!**



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Example, Bond Yields

- Solving for the yield analytically is hard, so use the financial calculator
- A bond's annual coupon rate is 5%. Its face value is \$1,000. Its maturity is 3 years. Its current market price is \$1,027.75. What is its yield?

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  - **CPT**, then **I/Y**

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  - **CPT**, then **I/Y**
- The yield is 4%

Required Rate of Return vs. Yield

Required rate of return r

- The discount rate for computing the bond's value
- Chosen by investors as the opportunity cost, the return on alternatives with a similar risk
- Input r, output the value (**CPT** **PV**)

Yield y

Required Rate of Return vs. Yield

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Yield y

- The discount rate for which the present value of the payments equals the price
- Implied by the market price
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Required Rate of Return vs. Yield

Required rate of return r

- The discount rate for computing the bond's value
- Chosen by investors as the opportunity cost, the return on alternatives with a similar risk
- Input r, output the value (CPT PV)
- **Subjective**

Yield y

- The discount rate for which the present value of the payments equals the price
- Implied by the market price
- Input the price, output the yield (CPT I/Y)
- **Objective**

Practice Problem

- The table shows data for three bonds. Each bond's coupon is zero. Each bond's face value is $1,000

Bond	Price	Maturity (years)	Yield
A	$310	25	?
B	$310	?	9%
C	?	11	8%

a. What is the yield of bond A?

b. What is the maturity of bond B?

c. What is the price of bond C?

Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

	N	I/Y	PV	PMT	FV
a. Bond A	25	?	−310	0	1,000
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Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

	N	I/Y	PV	PMT	FV
a. Bond A	25	4.796	−310	0	1,000
b. Bond B	?	9	−310	0	1,000
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- a. The yield of bond A is 4.796%

Practice Problem

- Solution, on the financial calculator (the highlighted key is the one solved for),

	N	I/Y	PV	PMT	FV
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c. Bond C	11	8	?	0	−1,000

- a. The yield of bond A is 4.796%
- b. The maturity of bond B is 13.59 years

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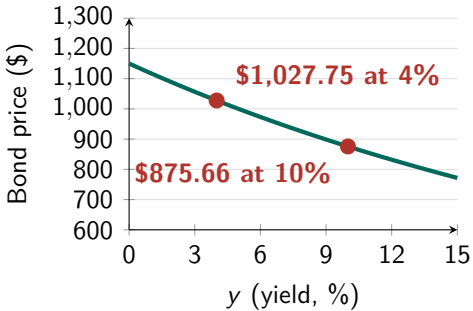
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a. Bond A	25	4.796	−310	0	1,000
b. Bond B	13.59	9	−310	0	1,000
c. Bond C	11	8	428.88	0	−1,000

- a. The yield of bond A is 4.796%
- b. The maturity of bond B is 13.59 years
- c. The price of bond C is $428.88

Bond Prices and Yields Move in Opposite Directions

- Because the bond yield is the y such that,

$$\text{Price} = \frac{\text{Coupon}}{(1+y)^1} + \frac{\text{Coupon}}{(1+y)^2} + \cdots + \frac{\text{Coupon} + \text{Face value}}{(1+y)^T}$$

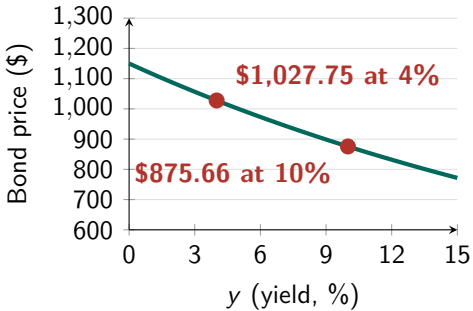


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- Higher y, lower bond price (the chart shows the running example bond)

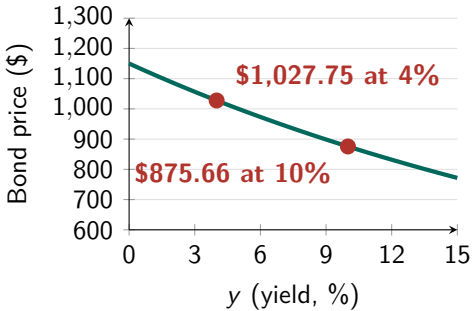


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What Determines Bond Price (and Therefore Yield)?

1. Market (risk-free) interest rates

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2. Risk of default
   - The risk that a bond issuer may default on its bonds
   - **Default premium**, the additional yield on a bond that investors require for bearing default risk

What Determines Bond Price (and Therefore Yield)?

1. Market (risk-free) interest rates
2. Risk of default
   - The risk that a bond issuer may default on its bonds
   - **Default premium**, the additional yield on a bond that investors require for bearing default risk

$$\textbf{Bond yield} = \textbf{risk-free rate} + \textbf{default premium}$$

Two Borrowers, Two Yields

- Borrower, the federal government
- Default risk, none
- Bond yield = risk-free rate
- Yield is low and price is high



Two Borrowers, Two Yields

- Borrower, the federal government
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- Borrower, a small corporation with risky cash flows
- Default risk, high
- Bond yield = risk-free rate + default premium
- Yield is high and price is low



Default Risk, Credit Ratings

- Credit rating agencies provide ratings intended to capture default risk
  - Largest, Moody's, Standard & Poor's, and Fitch

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- **Investment-grade bonds**, rated Baa or higher by Moody's, or BBB or higher by Standard & Poor's
- **High-yield bonds** ("junk" bonds), rated Ba or BB and lower

Default Risk, Credit Ratings

Moody's	S&P	Default within 10 yr
Investment-grade bonds		
Aaa	AAA	0.1%
Aa	AA	0.7%
A	A	2.2%
Baa	BBB	3.5%
High-yield bonds		
Ba	BB	15.7%
B	B	35.5%
Caa, Ca	CCC, CC	48.9%
C	C	n/a

Source, Brealey, Myers, and Marcus, Table 6.2

Default Risk, Credit Ratings

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- Yields tend to be higher (prices lower) for lower-rated bonds

Treasury Yields at 20-Year Highs

- **Treasury yield**, the yield on bonds issued by the U.S. government ("Treasury bonds")



PBS NewsHour, August 20, 2026 (7:14), "Why Treasury yields are at 20-year highs"

Recap of Part 2

- Yield, the discount rate for which the present value of the bond's payments equals its price

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Recap of Part 2

- Yield, the discount rate for which the present value of the bond's payments equals its price
- Bond prices and yields move in opposite directions
- Prices fall and yields rise with market interest rates and with default risk, lower-rated bonds carry higher yields

Module 5, Part 3

Stock Valuation, Dividend Discount Model

Valuing Stocks

- Valuing stocks helps us decide what to invest in

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- Two ways to value stocks,
  1. Dividend discount model
  2. Valuation by comparables

Valuing Stocks

- Valuing stocks helps us decide what to invest in
- Two ways to value stocks,
  1. Dividend discount model
  2. Valuation by comparables
- The approaches are complementary, no approach provides "the" right answer
- Professionals use a combination

Stock Valuation, Terminology

- **Dividend**, periodic cash paid out to shareholders

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- **Earnings per share (EPS)**, $\dfrac{\text{Net income}}{\text{Number of shares outstanding}}$

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- **Dividend**, periodic cash paid out to shareholders
- **Earnings per share (EPS)**, $\dfrac{\text{Net income}}{\text{Number of shares outstanding}}$
- **P/E ratio**, $\dfrac{\text{Stock price}}{\text{Earnings per share}}$

Valuing Stocks, Dividend Discount Model

- Suppose you plan to buy a stock and own it forever
- What price should you pay for the stock today (P)?

Valuing Stocks, Dividend Discount Model

- Suppose you plan to buy a stock and own it forever
- What price should you pay for the stock today (P)?
- Stock value, the present value of the cash flows you get from owning it,

$$P = \text{PV(Dividends)} = \frac{\text{Div}_1}{(1+r_e)^1} + \frac{\text{Div}_2}{(1+r_e)^2} + \frac{\text{Div}_3}{(1+r_e)^3} + \cdots + \frac{\text{Div}_\infty}{(1+r_e)^\infty}$$

Valuing Stocks, Dividend Discount Model

- Suppose you plan to buy a stock and own it forever
- What price should you pay for the stock today (P)?
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- **Required return on equity** (r_e), the discount rate, set by the opportunity cost of investing in the stock, the return on alternatives with a similar risk

Valuing Stocks, Dividend Discount Model

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- Div_1, Div_2, ... are *expected* dividends

Gordon Growth Model

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- The second formula is often called the "**Gordon growth model**"

Example, Gordon Growth Model

- 1. A firm will pay a $3 dividend forever (the first dividend exactly a year from now). Its required return on equity is 12%. What will investors pay for the stock?
- 2. What if the firm's stock pays a $3 dividend a year from now, and then the dividend grows at 8% forever?

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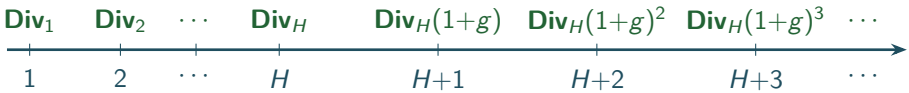
$$1. \quad P = \frac{\text{Div}_1}{r_e} = \frac{\$3}{0.12} = \$25$$

$$2. \quad P = \frac{\text{Div}_1}{r_e - g} = \frac{\$3}{0.12 - 0.08} = \$75$$

Valuing Dividends with Non-Constant Growth

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Valuing Dividends with Non-Constant Growth



Valuing Dividends with Non-Constant Growth

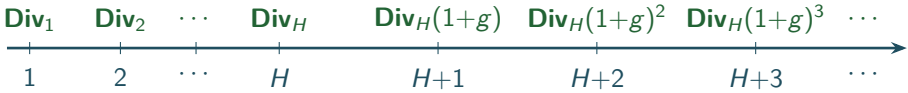
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$$\text{PV} = \frac{\text{Div}_1}{(1+r_e)^1} + \frac{\text{Div}_2}{(1+r_e)^2} + \cdots + \frac{\text{Div}_H}{(1+r_e)^H} + \frac{P_H}{(1+r_e)^H}$$

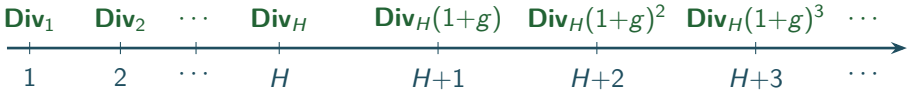
P_H ← Terminal value

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- Why? In practice, near-term dividends can be estimated carefully one year at a time, the distant future cannot

Example, Valuing Non-Constant Growth

- A stock's expected dividends are in the table. Its required return on equity is 10%. Its dividends grow at 5% a year beginning in year 6. What is the value of the stock?

Year	1	2	3	4	5
Dividend	$2	$2	$2	$3	$3

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$$\text{PV}_{\text{Stock}} = 39.12 + 8.89 = \$48.00$$

Practice Problem

- Consider the following two stocks,

a. Stock A is expected to pay a dividend of \$10 a share forever
b. Stock B is expected to pay a dividend of \$5 next year. Thereafter, dividend growth is expected to be 4% a year forever

- Each stock's required return on equity is 10%. Which stock is the most valuable? What if the required return on equity is 7%?

Practice Problem

Stock A



Practice Problem



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- Solution, both stocks are perpetuities, apply the perpetuity formulas,

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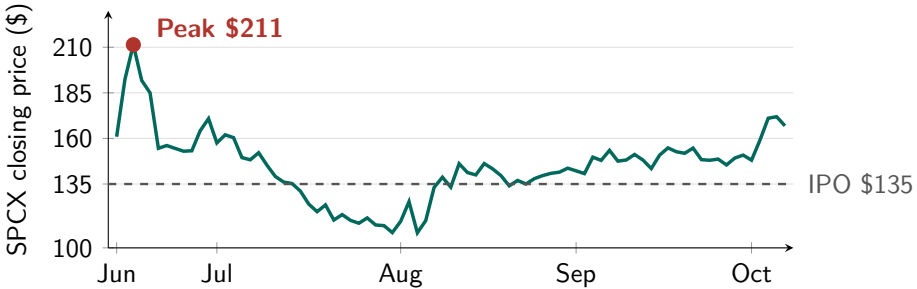
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- Now Stock B is the most valuable

Let's Hear from the "Dean of Valuation"



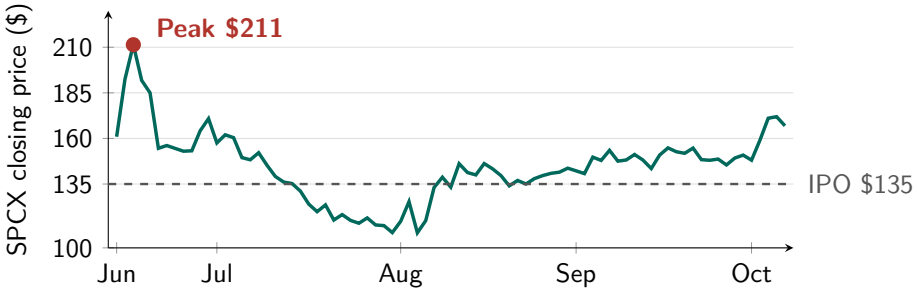
CNBC, June 12, 2026 (2:27), "SpaceX's valuation looks 'richly priced,' says NYU's Aswath Damodaran"

SpaceX Since the IPO



Source, Yahoo Finance, SPCX daily closing prices

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- **People who bought at the peak ($211) would regret not listening to Damodaran!**
- For more details on how Damodaran thinks of SpaceX valuation, see this (45:19)

Recap of Part 3

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- Constant dividends, $P = \text{Div}_1/r_e$, constant growth (Gordon growth model), $P = \text{Div}_1/(r_e - g)$

Module 5, Part 4

Stock Valuation, Comparables

Valuing Stocks, Comparables

- What is one of the most common approaches to valuing real estate?



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- Comparables (“comps”)!



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- What is one of the most common approaches to valuing real estate?
- Comparables ("comps")!
- E.g., a 2,000 sqft house in New Orleans sells for $400,000. What is your estimate for a neighboring house of 2,500 sqft?



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- E.g., Company A has twice the earnings per share of Company B, and Company B trades at $10. What price should Company A's shares trade at?
- Underlying idea, we can "scale" prices!

Valuing Stocks by Comparables, Four Steps

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3. Calculate the average of that multiple for the comparables **A**, **B**, **C**, ...
4. Back out Company **X**'s implied value as the average multiple times Company **X**'s denominator

$$P_{\mathbf{X}} = (\text{Average P/E of comparables } \mathbf{A}, \mathbf{B}, \mathbf{C}) \times \text{EPS}_{\mathbf{X}}$$

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- To get the price per share,

$$P = \frac{\text{Market capitalization}}{\text{Number of shares outstanding}} = \frac{\$30 \text{ billion}}{1 \text{ billion}} = \$30$$

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- Two ways to value stocks, the dividend discount model and comparables

Thank you!

Please let me know if you have any questions.